# EME OSPE

Station 1

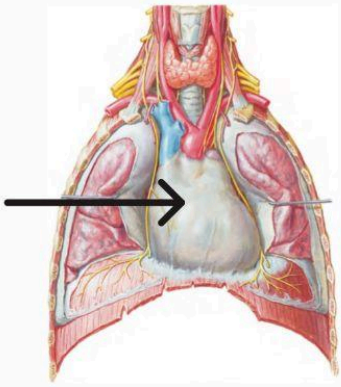


Figure 1

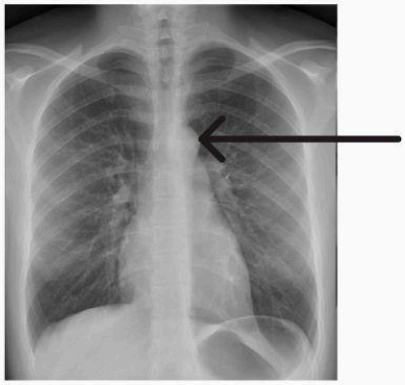


Figure 2

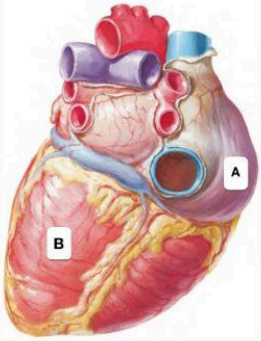



Figure 3

1. Name the appointed structure in Figure 1. What is its blood and nerve supply? How does it support the organ that it encloses?
    - Fibrous pericardium
    - Blood supply: pericardiocophrenic arteries and veins
    - Nerve supply: phrenic nerve
    - Function: prevents overdistension of the heart; its base is adherent to the central tendon of the diaphragm to support the heart during contraction

2. What is the structure in Figure 2? What is it formed by?
    - Aortic knuckle
    - Formed by the lateral edge of the distal aortic arch

3. Identify structures A and B in Figure 3 and their development
    - A: RA
        - Sinus venarum (smooth part) from sinus venosis
        - Pectinate muscle (rough part) from the primitive atrial chamber
    - B: LV
        - Aortic vestibule (smooth part) from bulbous cordis
        - Trabecula carnae (rough part) from the primitive ventricular chamber

Station 2

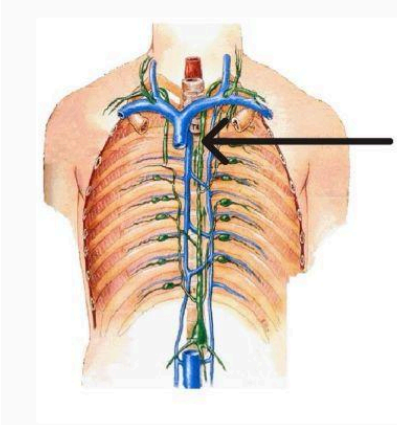


Figure 1

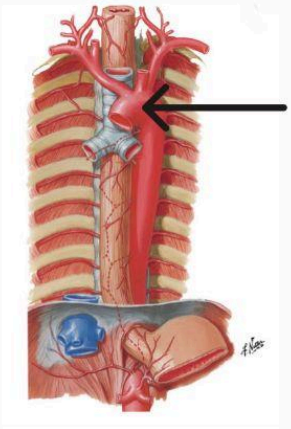


Figure 2

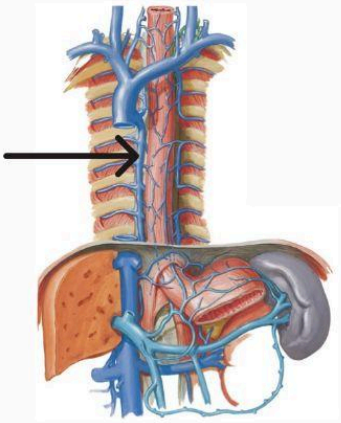


Figure 3

1. Identify the appointed structure (green color) in Figure 1. Where does it terminate?
    - Thoracic duct
    - Drains into the junction between the left subclavian vein and the left internal jugular vein

2. Identify the structure in Figure 2, name three of its branches
    - Aortic arch
    - Left subclavian artery, left common carotid artery, brachiocephalic trunk

3. Identify the appointed structure (blue color) in Figure 3. Name two of its tributaries.
    - Azygous vein system
    - Superior and inferior hemiazygous veins

Station 3

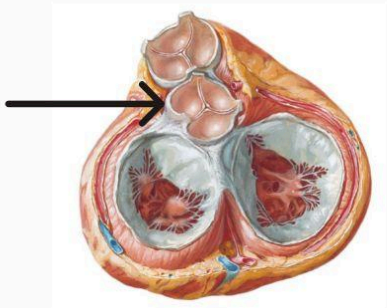


Figure 1

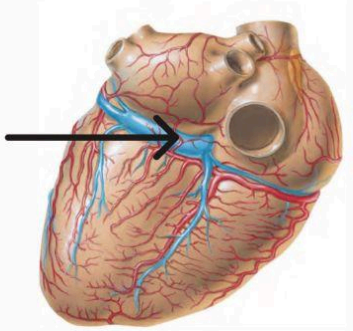


Figure 2

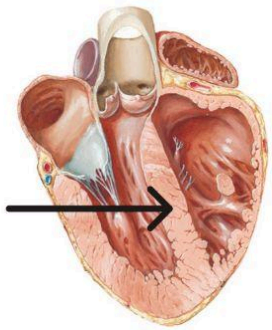


Figure 3

1. What is the arrow pointing to in Figure 1? Name two of its branches and one clinical correlation
    - (Incorrect answer) Left coronary artery - LAD and PDA - occlusion leading to ischemia of the myocardial tissue and requiring a surgical bypass
    - (Confirmed answer) Ascending aorta - RCA and LCA - ligation (transverse pericardial sinus) in open heart surgery

2. What is the structure in Figure 2? Name two of its tributaries. Where does it drain?
    - Coronary sinus
    - The great cardiac vein and the middle cardiac vein
    - The coronary sinus terminates in the right atrium

3. What is the structure in Figure 3? What are its parts? Name its blood supply
    - Interventricular septum
    - Muscular part and membranous part
    - Anterior ⅔ supplied by LAD, posterior ⅓ supplied by PDA

Station 4 (MCQs)

A 28-year-old patient goes to his routine check-up. His cholesterol levels are around 300 mg/dL. His father died suddenly at the age of 45. He had a persistently high cholesterol level his entire life. The doctor prescribes a lipid-lowering agent, and in the next levels, the patient's cholesterol is lowered significantly.

What is the most likely diagnosis of this patient?

| | |
|---|---|
| Familial hyperalphalipoprotein deficiency | |
| Type IIa familial hypercholesterolemia | ✓ |

What is the cause of this disorder?

| | |
|---|---|
| VLDL overproduction | |
| LDL receptor defect | ✓ |

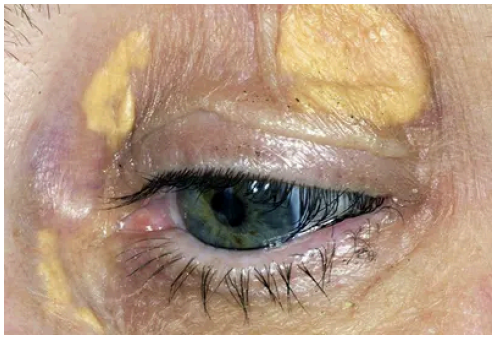
What is a common risk factor of this disorder?

| | |
|---|---|
| Premature atherosclerosis | ✓ |
| Fat malabsorption | |

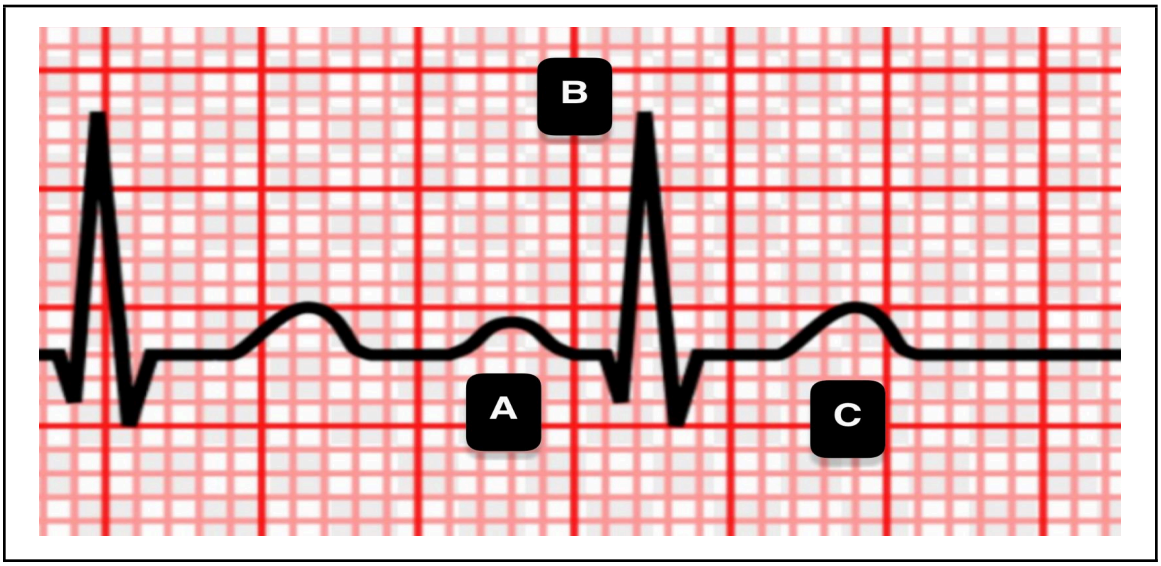
Which lipoprotein will most likely be within normal range in this disease? (confirmed)

| | |
|---|---|
| LDL | |
| HDL | |
| Triglycerides | ✓ |

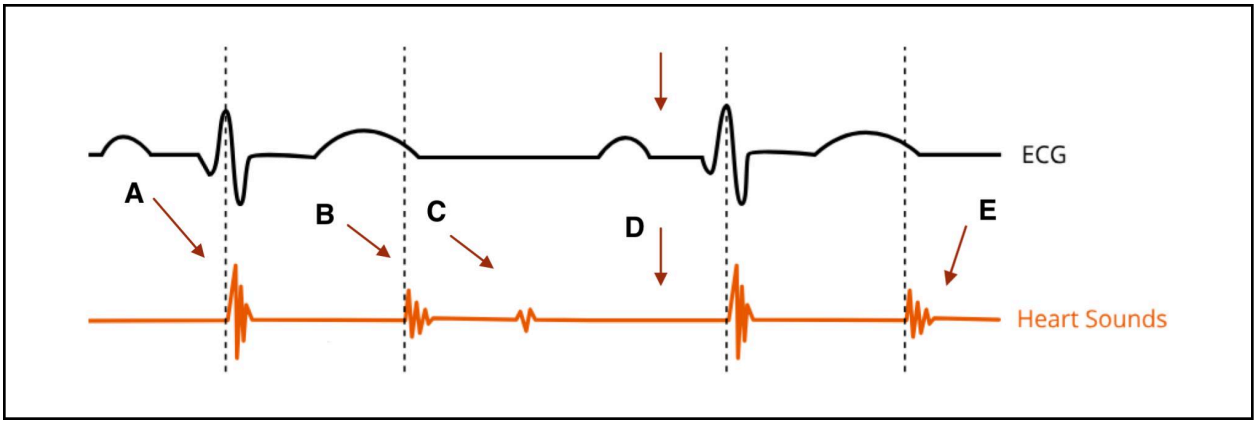
What is this lesion that is commonly associated with the pateinst disorder?

| | | |
|---|---|---|
|  | Xanthoma | ✓ |
| | Phrynoderma | |

Station 5




1. Mention what the letters represent and their corresponding electrical activity from the attached ECG strip
    - A: P wave, atrial depolarization
    - B: QRS complex, ventricular depolarization
    - C: T wave, ventricular repolarization
2. Calculate the heart rate from the attached ECG strip
    - 1500/17 small boxes = 88bpm (normal sinus rhythm)
3. Mention two indications (uses) for ECG
    - Diagnose arrhythmias
    - Detect electrolyte imbalances




Q. Identify the heart sounds; one sound can be used more than once

A. A: S1 - B: S2 - C: S3 - D: S4 - E: S2

Station 6

A 52-year-old patient's lipid profile shows serum cholesterol of 280 mg/dl, LDL of 151 mg/dl, and HDL of 37 mg/dl.

1. What's your working diagnosis?
    - Tab. Atorvastatin 10 mg once a day at night for 1 month
      Review after one month
2. Why did you choose these therapeutic options for this patient?
    - Atorvastatin is a potent statin known for effectively lowering LDL cholesterol levels. Statins are not only effective at reducing cholesterol but also significantly lower the risk of cardiovascular events (such as heart attacks and strokes) in patients with dyslipidemia

Station 7

A patient comes with fever and a new regurgitant murmur; blood culture is negative.

1. What is your diagnosis?
    - Culture-negative infective endocarditis
2. What are two common causes of this patient's condition?
    - Prior antibiotic use
    - Infection of damaged valves with fastidious organisms such as HACEK group or Coxiella burnetii (an obligate intracellular organism)





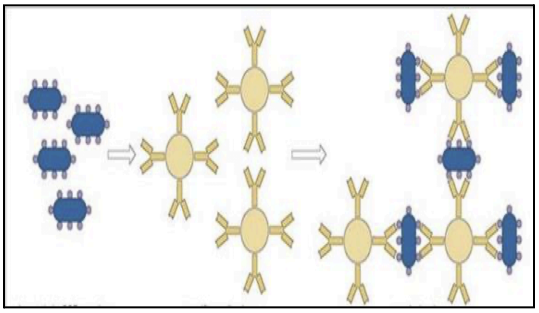



Q. Based on the following laboratory findings, microscopic features, and special test, identify the organism causing this patient's infection (this question is independent from the first case Scenario)

A. Enterococcus faecalis
    - Oval-shaped gram-positive cocci in pairs
    - Non-hemolytic translucent colonies on blood agar
    - Positive bile esculin test

Station 8

Case Scenario for a patient with rheumatic heart disease

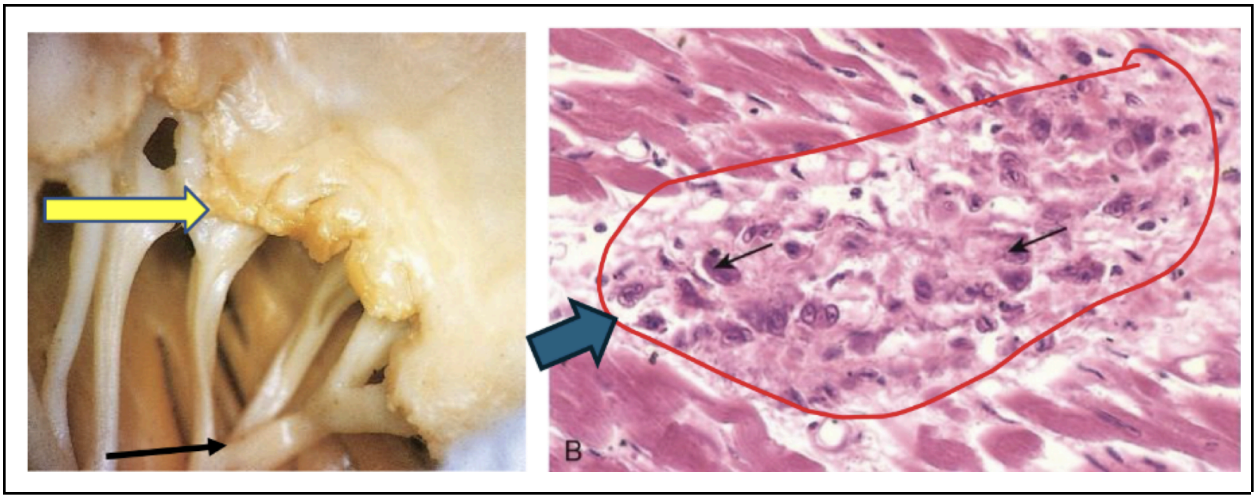


1. Mention nonspecific markers associated with RHD
    - ESR and CRP
2. What is the importance of the ASO test in RHD, and what value is clinically significant?
    - Antistreptolysin O titre (ASOT) determines antibodies against SLO produced by the immune system in response to the streptococcal infection
    - ASO titer >200 IU is considered a positive test
3. What test is used in diagnosing RHD based on the attached image? Is it based on agglutination or precipitation?
    - Latex agglutination test
    - Agglutination
4. Co-agglutination participates in agglutination by binding the antibody to which of the following?

| | |
|---|---|
| A) Nucleated avian RBCs | |
| B) Latex particles | |
| C) Staphylococcus aureus (Cowan I strain) cells | ✓ |

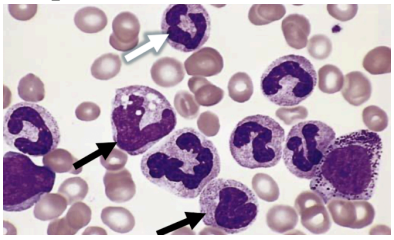
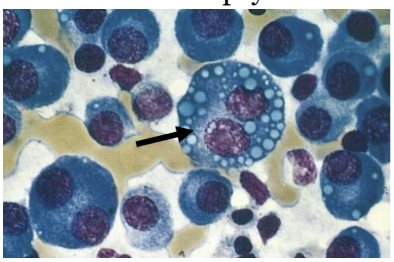
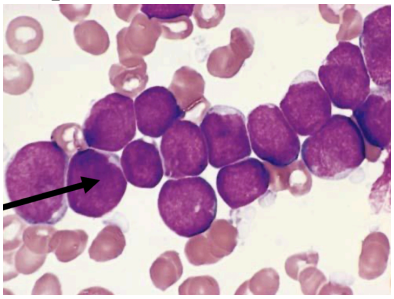
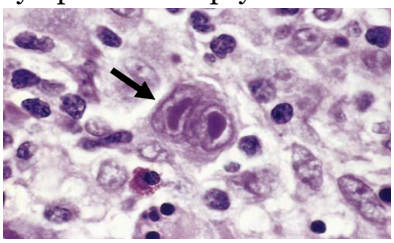
Station 9

(Case Scenario)




1. Mention one gross and one microscopic feature
    - Gross: small vegetations along the line of closure of the mitral valve leaflet
    - Microscopic: a circumscribed nodule of mixed mononuclear inflammatory cells with fibrinoid necrosis (aschoff body)
2. What is your diagnosis? (3 marks)
    - Rheumatic heart disease

Station 10 (Match)

| | | |
|---|---|---|
| Peripheral blood smear  | | Lymphoblasts are shown with condensed stippled nuclear chromatin. Small nucleoli. Scant agranular basophilic cytoplasm |
| Bone marrow biopsy  | | Plasma cells with atypical forms with multiple nuclei, cytoplasmic droplets containing immunoglobulin (Russell bodies) |
| Peripheral blood smear  | | Reed Sternberg cell with two mirror image nuclei or nuclear lobes, “owleye” appearance |
| Lymph node biopsy  | | Neoplastic cells comprise both mature forms (neutrophils, white arrow) & immature granulocytic forms in various stages of differentiation |